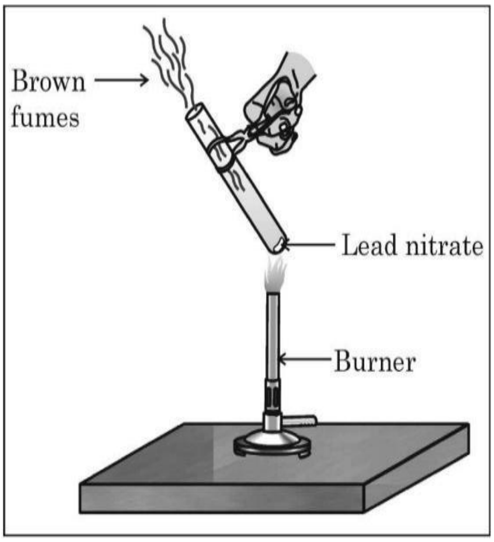
KENDRIYA VIDYALAYA SANGATHAN ,ERNAKULAM REGION
PRE BOARD EXAMINATION (2024-25)

CLASS X **SUBJECT: SCIENCE(086)** **MAX.M:80** **TIME:3h**

General Instructions:
1. All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.
2. Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.
3. Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.
4. Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.
5. Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.
6. Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1/2/3 marks.

Section-A

Question 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions

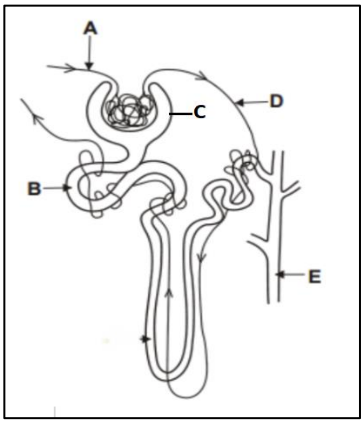
1	Meera performed the experiment given below using lead nitrate and observed the emission of brown fumes. This is due to  A. Oxidation of lead nitrate to produce lead oxide and oxygen B. Displacement of lead C. Decomposition of lead nitrate to produce nitrogen dioxide D. Double displacement reaction .	1
2	Identify ‘a’,‘b’and ‘c’in the following balanced reaction. **a** Fe (s)+ **b** H_2O(g) → Fe_3O_4 (s) + **c** H_2(g) A.4,3,4 B.3,4,4 C. 2,3,3 D.4,4,3	1
3	Calcium oxide reacts vigorously with water to produce calcium hydroxide.This reaction can be classified as (a)combination (b)oxidation (c)Exothermic (d)endothermic A.(a) and (b) B.(b) and (c) C.(a) and (c) D.(a) and (d)	1
4	Which of the following will form alkalies? (a)K_2O (b)SO_2 (c)Na_2O (d)NO_2 A.(a) and (b) B.(a) and (c) C.(b) and (c) D.(a) and (d)	1
5	Reena was preparing HCl gas from solid NaCl and concentrated H_2SO_4 and testing the nature of the gas. Teacher advised her to pass the HCl gas produced through a guard tube containing calcium	1

	chloride for better results. The role of calcium chloride is to A. moisten the gas evolved B. absorb the gas evolved C. absorb moisture from the gas D. absorb the Cl^- ions from the gas.	
6	Samples of four metals A, B, C and D were taken and added to the following solution one by one. The results obtained have been tabulated as follows	1

Metal	Iron(II)sulphate	Copper(II)sulphate	Zinc sulphate	Silver nitrate
A	No reaction	displacement		
B	displacement		No reaction	
C	No reaction	No reaction	No reaction	Displacement
D	No reaction	No reaction	No reaction	No reaction

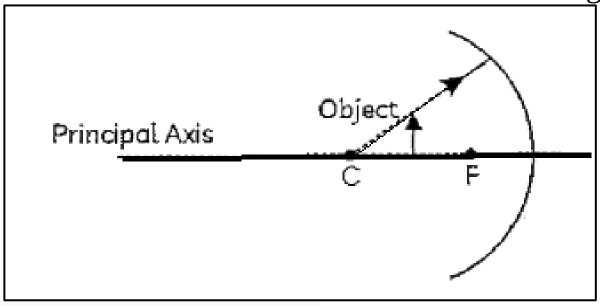
Use the table above and arrange the metals A,B,C and D in the order of decreasing reactivity
A. B >A > C >D
B. A> B> C > D
C. D> C > B> A
D. C > A > B >D

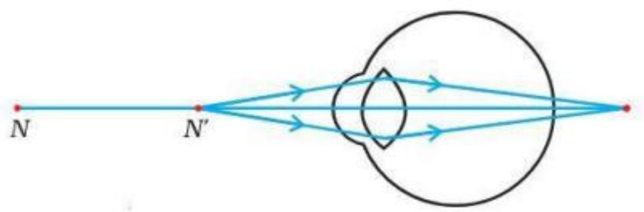
7	An alloy is a homogeneous mixture of two or more metals, or a metal and a nonmetal. Bronze is an alloy of A. copper and zinc B. copper and tin C. lead and tin D. lead and zinc	1
8	The nephron is the functional unit of the kidney and nephron's primary function is to filter the blood, reabsorb needed substances, and remove wastes.The part of nephron where selective re absorption occurs is	1

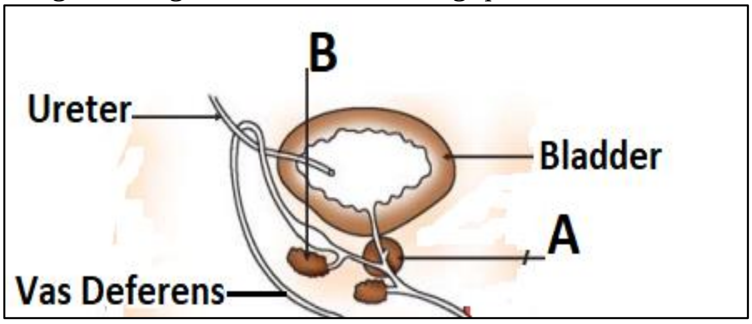


A. C B.A C. E D. B

9	The heart is a pump, usually beating about 60 to 100 times per minute and with each heartbeat, the heart sends blood throughout our bodies, carrying oxygen to every cell. Identify the process in the heart that leads to the pumping of oxygenated blood to all parts of the body? A. contraction of the left atrium B. contraction of left ventricle C. relaxation of the right atrium D. relaxation of the right ventricle	1
10	Bile is a yellowish-green digestive fluid produced by the liver and stored by the gallbladder.Which part of alimentary canal receives bile from the liver? A.stomach B.small intestine C.large intestine D.oesophagus	1
11	The part of the human brain that is responsible for the precision of voluntary actions and maintaining the posture and balance of the body. A. cerebrum B. medulla C. cerebellum D. pons	1
12	A pea plant with round green seed (RRyy) is crossed with another pea plant with wrinkled yellow seed(rrYY).What would be the nature of seeds in the first generation? A. Round Green B.Round Yellow C.Wrinkled Green D.Wrinkled Yellow	1

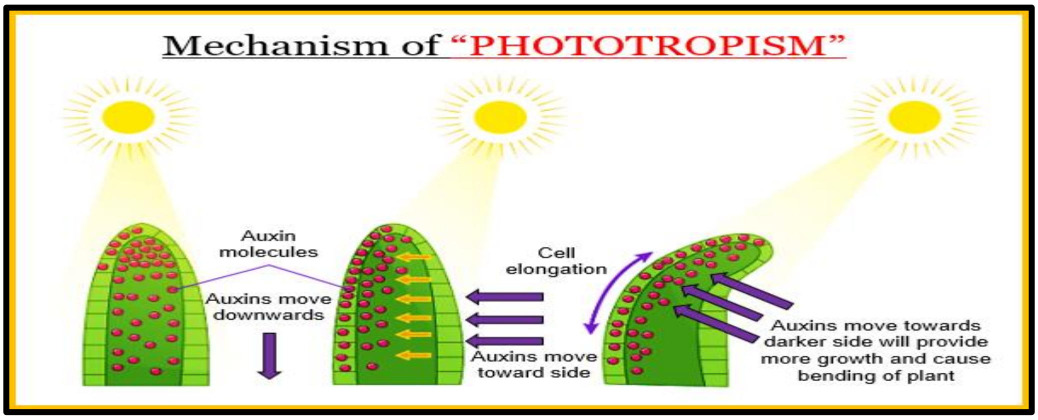
13	Identify the correct conclusions that can be drawn based on the diagram of the concave mirror given below .  I) the image of the object will be inverted II) the focal length will be negative III) the image of the object will be virtual IV) the reflected ray will travel along the same path as the incident ray in the opposite direction A.I and III B.II and III C.III and IV D.I,II and IV	1
14	The focal length of the eye lens decreases when eye muscles A.are relaxed and lens become thinner. B. contracts and lens becomes thicker C.are relaxed and lens become thicker D.contracts and lens become thinner	1
15	In a food chain if frog is eaten by snake then the energy transfer will be from A. producer to primary consumer B.primary consumer to secondary consumer C.secondary consumer to tertiary consumer D.Producer to decomposer	1
16	In the following food chain, plants provide 300 J of energy to rats. How much energy will be available to hawks from snakes? Plants → Rats → Snakes → Hawks A.300J B.30J C.3J D.0.3J	1
Question No. 17 to 20 consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true		
17	Assertion (A):Plants produced by vegetative propagation are exact copies of their parents Reason (R):In vegetative propagation, off springs are produced by the fusion of gametes.	1
18	Assertion (A):Light slows down if the refractive index of a medium increases. Assertion (R):The speed of light in a medium is inversely proportional to the refractive index of the medium	1
19	Assertion(A):Biodegradable and non biodegradable waste should be discarded separately Reason (R):Biodegradable wastes are not harmful	1
20	Assertion (A):Sodium chloride is an acidic salt. Reason (R):Salts of a strong acid and weak base are acidic.	1
Section-B Question No. 21 to 26 are very short answer questions		
21	(a) Identify the substance oxidised and the substance reduced in the reaction given below. $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$. (b) Translate the statement given into chemical equation and balance it. Barium chloride reacts with aluminium sulphate to give aluminium chloride and a precipitate of barium sulphate	2
22	Sphincters are ring-like muscles that play many vital roles in the human body.Name any two sphincter muscles associated with our alimentary canal and write about the function of each. OR Green plants capture light energy and use this light energy to convert water, carbon dioxide, and minerals into oxygen and energy-rich organic compounds like glucose.List the steps involved in the synthesis of glucose by the plants	2
23	The placenta develops shortly after conception and attaches to the wall of the uterus. What is placenta?	2

	Mention its role during pregnancy.	
24	An object is kept at a distance of 15cm,20cm and 22cm respectively from a lens of power+5D. a) Identify the type of lens giving reason for your identification. b) In which case/cases would you get i) a virtual,erect and magnified image ii) real,inverted and magnified image	2
25	Joule’s law is a mathematical description of the rate at which resistance in a circuit converts electric energy into heat energy. a) State Joules law of heating. b) Write any two applications of this law from your daily life OR a) What are the advantages of connecting electrical devices in parallel with the battery instead of connecting them in series.(any two) b)Why is an ammeter connected in series in an electric circuit?	2
26	A gas Y made up of three oxygen atoms (O_3)occurs naturally in small amounts in the upper atmosphere (the stratosphere). a) How is this gas formed at the higher levels of atmosphere? b) “Damage to this layer of gas Y in the upper atmosphere is a cause of concern” .Justify this statement.	2
	Section-C Question No. 27 to 33 are short answer questions	
27	A white substance X is commonly used for supporting fractured bones in the right position. a)Identify X and write its chemical name b)Explain the preparation of the given compound X with the chemical equation involved. c)Why should you always store X in moisture proof containers?	3
28	Refining of metals is the process of purifying impure metals. Explain electrolytic refining of copper with the help of a neat labelled diagram. OR Some metal oxides like aluminium oxide and zinc oxide are amphoteric in nature a) What are amphoteric oxides? b) Justify your definition with the help of relevant equations.	3
29	Cellular respiration is a series of chemical reactions that break down glucose to produce ATP. Draw a schematic diagram representing the different ways by which glucose is broken down to release energy.	3
30	a) In human beings, male has 46 chromosomes and female has 46 chromosomes. The offspring produced by the fusion of male and female gametes also has 46 chromosomes and not 46 pairs of chromosomes. Justify the statements by giving possible reason. b)With the help of a flow chart, explain sex determination in a human baby, if the sperm carrying the X chromosome fertilizes the egg.	3
31	a) Draw the pattern of magnetic field lines of the magnetic field produced by a solenoid through which a steady current flows. b)What does the field pattern inside the solenoid indicate?	3
32	 Observe the diagram and answer the following questions: (a) Name the defect of vision shown in figure. What will be the optical experience of the person suffering from this defect? (b) Give two possible reasons for this defect of eye in human beings (c) Draw a labelled diagram to show how the defect is rectified by using suitable lens.	3
33	Draw a schematic diagram of a circuit consisting of a battery of 3 cells of 2V each,a 5Ω,8Ω and a 12Ω resistor connected in series ,an ammeter to measure the current through resistors and a voltmeter to	3

	measure the potential difference across the 12Ω resistor and a plug key.What would be the reading in the ammeter and voltmeter?	
	Section-D Question No. 34 to 36 are long answer questions.	
34	An organic compound A of molecular formula$C_2H_4O_2$ is widely used as preservative in pickles. This compound reacts with ethanol to form a sweet smelling compound B. (a)Identify compound A and B. (b)Write the chemical equation for the reaction involved. (c)Write two uses of compound B. (d)What happens when compound A reacts with baking soda? Write chemical equation for the reaction. (e)What happens when compound A is treated with sodium hydroxide? Write chemical equation for the reaction. OR (a)What are micelles? (b)With the help of suitable diagram show the formation of micelles,when soap is applied to dirt. (c)Explain the mechanism of cleansing action of soap . (d)What is saponification reaction? Write the equation involved.	5
35	Study the given diagram and the following questions  a)Identify A and B that are vital parts of human male reproductive system and write two functions of these parts. b)Suggest three contraceptive methods that can be practiced to control the size of human population which is essential for the health and prosperity of the country.State the basic principle involved in each. OR After pollination, when pollen has landed on the stigma of a suitable flower of the same species, a chain of events happen.Describe post pollination changes with the help of a neat labeled diagram showing germination of pollen on the stigma.	5
36	a) State the law that gives the relationship between current through a conductor and potential difference across its two terminals. b) How is this law verified experimentally? c) The potential difference between the terminals of an electric heater is 60 V when it draws a current of 4 A from the source. What current will the heater draw if the potential difference is increased to 120 V? d) A piece of wire of resistance R is cut into five equal parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R′, then find the ratio of R/R′ ? OR a) What is resistance of a conductor? b) On what factors does the resistance of a conductor depend? c) State the relationship between the resistance R of a wire to its length l and cross-sectional area A. Use the mathematical symbols to arrive at the final formula. Deduce the formula to find resistivity from the final formula d) A copper wire has diameter 0.5 mm and resistivity of 1.6×10^{-8} Ω m. What will be the length of this wire to make its resistance 10 Ω?	5
	Section – E Question No. 37 to 39 are case-based/data -based questions.	

37 — **AUXINS AND PHOTOTROPISM** — 4

Auxins are a family of plant hormones. They are mostly made in the tips of the growing stems and roots, which are known as apical meristems, and can diffuse to other parts of the stems or roots.Phototropism is caused by an unequal distribution of auxin.



A. Name a phytohormone which inhibits growth in plants.(1)

B. Give one example for chemotropism.(1)

Attempt either subpart C or D

C. Differentiate between the response shown by the leaves of a sensitive plant on being touched and the growth of a tendril around a support.(2)

OR

D. Relate the action of auxin at the shoot tip to the curving of plant towards light source.

38 — **HYDROCARBONS** — 4

Hydrocarbons are organic molecules made up of carbon and hydrogen atoms that are held together by covalent bonds . Even though they are composed of only two types of atoms, there is a wide variety of hydrocarbons because they may consist of varying lengths of chains, branched chains, and rings of carbon atoms, or combinations of these structures. In addition, hydrocarbons may differ in the types of carbon-carbon bonds present in their molecules.

A. Carbon shares its valence electrons with other carbon atoms or with atoms of other elements rather than gaining or losing valence electrons. Give reason to justify this statement (2)

B. List two properties of compounds formed by the sharing of valence electrons.(1)

Attempt either sub part C or D.

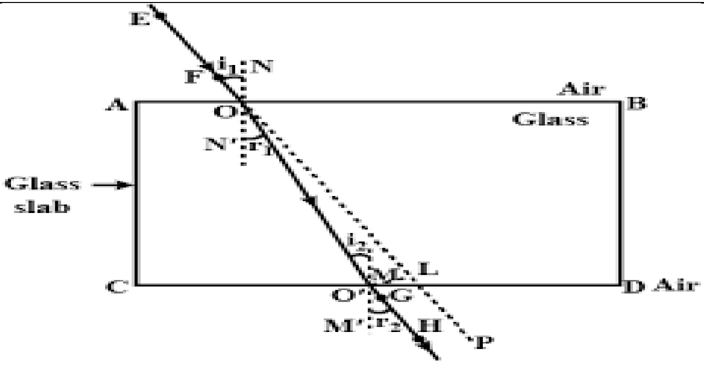
C. Write the structural formula of i)Hexanal and ii)Propanone(1)

OR

D .Draw the electron dot structure of Ethyne

39 — **REFRACTION** — 4

When electromagnetic radiation, in the form of visible light, travels from one substance or medium into another, the light waves may undergo a phenomenon known as refraction, which is manifested by a bending or change in direction of the light. Refraction occurs as light passes from one medium to another only when there is a difference in the index of refraction between the two materials. The effects of refraction are responsible for a variety of familiar phenomena, such as the apparent bending of an object that is partially submerged in water and the mirages observed on a hot, sandy desert.



A. A ray of light travelling in air enters obliquely into glass. Does the light ray bend towards the normal or away from the normal? Why?(1)

B. State Snell's law of refraction.(1)

Attempt either subpart C or D.

C. Refractive index of diamond with respect to glass is 1.6 and absolute refractive index of glass is 1.5. Find out the absolute refractive index of diamond.(2)

OR

D. Light enters from air to glass having refractive index 1.50. What is the speed of light in the glass? The speed of light in vacuum is 3×10^8 m/s.

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